Radiation, reservoirs and the action-scale question

A finite universal equilibrium radiation spectrum with Wien scaling and the Rayleigh–Jeans low-frequency normalization defines a positive action **unit**. It does not impose a minimum body action or a recording restriction. Linear radiation balance retains a freely supplied field amplitude; a nonlinear pumped oscillator has a genuine action attractor but its value depends on supplied power and frequency. Classical two-pointer records defeat a prior-only density ceiling when the specified instruments are available. A stipulated information– work inequality prices resolution without selecting a positive action floor.

These are separate results under separate physical assumptions. Physical $\hbar$ remains explicit: identifying a unit or reservoir parameter with it is an additional claim. This treatment integrates the radiation and energy-depot tests with the unit/indeterminacy distinction. Written model/coordinator reviews were internal, not independent peer review; the ingredients have the cited prior art and no novelty claim is made for their assembly.

A normalized radiation action unit

Let $u(\nu, T) \geq 0$ be a measurable equilibrium spectral energy density, $T > 0$, and assume:

- (K) the same spectrum for every material enclosure;
- (W) Wien scaling $u(\nu, T) = \nu^3 f(\nu/T)$;
- (S) $0 < \int_0^\infty u(\nu, T) d\nu < \infty$;
- (R) $u(\nu, T)/[(8\pi\nu^2/c^3)k_B T] \to 1$ as $\nu/T \to 0$, with fixed $c, k_B > 0$.

Theorem U. Put $x = \nu/T$ and define

$$r(x) = \frac{c^3 x f(x)}{8\pi k_B}, \quad I = \int_0^\infty x^2 r(x) dx, \quad \beta = I^{-1/3}, \quad h_{\mathrm{rad}} = k_B \beta.$$

Then $0 < I < \infty$, $[\beta] = \mathrm{K\,s}$ and $h_{\mathrm{rad}} > 0$ is an enclosure-independent action unit. With $\varphi(y) = r(y/\beta)$,

$$u(\nu, T) = \frac{8\pi k_B T \nu^2}{c^3} \varphi(\beta\nu/T), \quad \varphi(0+) = 1, \quad \int_0^\infty y^2 \varphi(y) dy = 1.$$

If $\int u \, d\nu = \sigma' T^4$, then

$$h_{\mathrm{rad}} = \left(\frac{8\pi k_B^4}{c^3 \sigma'} \right)^{1/3}.$$

Proof. Substitution gives $\int u \, d\nu = (8\pi k_B/c^3) T^4 I$. Nonnegativity and (S) make I finite; (R) makes it positive. Universality of f makes I independent of enclosure. Its units are $(\mathrm{K\,s})^{-3}$. Changing variable $y = \beta x$ proves the remaining identities. □

The normalization is conventional; fixed positive numerical multiples are equally universal units. Enclosure independence permits dependence on other universal constants, including microscopic masses or charges. It is not independence from every such constant. No finite-monomial hypothesis is used. If $y^4 \varphi(y)$ has a specified finite positive maximizer y_*, the wavelength peak $\lambda_{\max} T = b$ satisfies $h_{\mathrm{rad}} = y_* k_B b/c$. Integrability alone permits narrow high-frequency spikes and does not imply that peak or its uniqueness.

For the Planck shape $\varphi_P(y) = y/(e^y - 1)$ in its usual normalization, $\int y^2 \varphi_P dy = \pi^4/15$. Thus

$$h_P = (\pi^4/15)^{1/3} h_{\mathrm{rad}}, \qquad h_P = 2\pi\hbar$$

when that measured shape is identified with physical Planck radiation. Its wavelength maximum has $y_{*,P} \simeq 4.965$. Sources: Wien 1893 reprint, metadata; Planck 1901, metadata. The earlier Wien-law entropy paper is not the source for the Planck distribution.

A second available unit is $k_e/c = \alpha\hbar$ for a fixed elementary Coulomb coupling. Dimensional and orbital selection proves its exact scope. With both units available a proposed floor may depend on their dimensionless ratio. Neutral composite probes may still be polarizable; they do not remove charge dependence. Neither an exponential spectral tail nor a choice of unit derives $p = h_P/\lambda$ as an individual corpuscle law.

Extending a radiation unit to other fields

The normalized radiation theorem above does not establish that every well-posed classical wave equation requires a universal quantum action. A proposed thermal extension would need a specified mode density, equilibrium state, actual equipartition and coupling to matter: infinitely many modes with a nonsummable k_BT contribution preclude finite equilibrium energy within that model. A cutoff, changed occupation law or different thermal premise changes the conclusion. Finite energy alone does not identify a unique occupation law or its constant across distinct fields.

Cross-field universality under mutual equilibrium remains a conjectural physical step. Einstein's solid-vibration use of Planck's constant 1907 (metadata) is a historical application, not a proof for every wave field. Deterministic PDE existence, a thermal state and a quantum continuum limit are different constructions. Superrenormalizable power counting alone is likewise not a convergence proof; four-dimensional running and the positivity of an infrared spectral coefficient require the separate gauge estimates.

A prior density ceiling does not close under classical records

Theorem I. Suppose a classical instrument class permits Hamiltonian body/pointer dynamics, the product rectangular preparations below, Bayesian conditioning on finite-resolution coordinate records, and the two specified impulsive couplings with storage of the first reading. There are no additional limits on squeezing, energy, gain or readout precision. For every $X, \eta > 0$ it contains a finite protocol whose final posterior support widths obey $\Delta q\,\Delta p < \eta$, although every initial canonical pair has density at most $1/X$.

Proof. Prepare the body uniformly on $[0, a] \times [0, X/a]$, pointer one (y, p_y) on $[0, \varepsilon] \times [0, X/\varepsilon]$, and pointer two (z, p_z) independently on $[0, \varepsilon_2] \times [0, X/\varepsilon_2]$. A pulse gqp_y, $g > 0$, gives $y \mapsto y + gq$, $p \mapsto p - gp_y$. A stored y bin of width ε' confines q to width $(\varepsilon + \varepsilon')/g$. A second pulse $g_2 p_y p_z$, $g_2 > 0$, sends $z \mapsto z + g_2 p_y$, $y \mapsto y + g_2 p_z$, leaving q, p, p_y fixed. Retain the earlier reading and read z to bin width ε_2'. Then

$$\Delta q\,\Delta p \le \frac{X(\varepsilon + \varepsilon')}{ag} + \frac{(\varepsilon + \varepsilon')(\varepsilon_2 + \varepsilon_2')}{g_2}.$$

Small positive $\varepsilon, \varepsilon'$ suffice at fixed finite other parameters. Both maps preserve phase volume, and each prior rectangle has area X. Unread conjugate ranges grow as coordinate widths shrink. □

This is a four-part instrument/preparation counterexample, not a claim that abstract Hamiltonian dynamics guarantees the required apparatus. A universal posterior restriction must exclude at least one of its assumptions. The regulated Gaussian bath treatment extends the two-pointer escape to every finite joint Gaussian covariance, including body–pointer correlations, and finite pulse durations at fixed cutoff.

Posterior area and the quantum disturbance bound are different: its variances concern specific hybrid states, not later conditional inference of a realized classical impulse. A posterior ceiling alone does not supply that statistical-speed theorem. Quantum kinematics, a restriction

on admissible readout/composition, and a correlated preparation class are different possible inputs, each needing its own physical premise. The covariance-restricted Gaussian theory of Bartlett–Rudolph–Spekkens (2012, author abstract) supplies more than a pointwise prior density ceiling, including instrument restrictions.

Nor does a bare Coulomb impulse supply indeterminacy. Integrating the force along an externally prescribed straight path (ut, b) gives $|\Delta p_\perp| b = 2k/u$. It is a small-deflection approximation for scattering. Exact repulsive Rutherford scattering instead has

$$b|\Delta p| = \frac{2k/u}{\sqrt{1 + [k/(\mu u^2 b)]^2}} \longrightarrow 0 \quad (b \downarrow 0),$$

with reduced mass μ. An infinitely heavy repulsive centre and $u \ll c$ give a nonrelativistic counterexample; finite particle size adds a new cutoff. A calculable impulse magnitude is not an unknown stochastic spread.

Stable regulated radiation balance: shape and free amplitude

In SI units supply a centred stationary **Gaussian** field, a laboratory cutoff $\Lambda > \omega_0 > 0$ and the stable weak-damping dipole approximation

$$m\ddot{x} + m\gamma\dot{x} + m\omega_0^2 x = qE(t), \qquad \gamma = \frac{q^2\omega_0^2}{6\pi\epsilon_0 m c^3} > 0,$$

$$\langle E(t)E(0)\rangle = \int_{\mathbb{R}} \frac{d\omega}{2\pi} S_E(\omega) e^{-i\omega t}, \qquad S_E(\omega) = \frac{K}{3\pi\epsilon_0 c^3} |\omega|^3 1_{|\omega|\le\Lambda}, \quad K \ge 0.$$

The damping is the cycle-averaged Larmor loss matched near resonance, $\gamma/\omega_0 \ll 1$; it is not a high-frequency radiation-reaction law. The cutoff selects a frame. The field, trap and dipole approximation are premises; Gaussian tails impose no hard speed ceiling. The ensemble action is $J = m\langle \dot{x}^2 + \omega_0^2 x^2\rangle/(2\omega_0)$. At fixed positive damping and finite cutoff, the homogeneous solution decays and Fourier response gives exactly in this effective model

$$J_* = \frac{q^2}{2m\omega_0} \int_{-\Lambda}^{\Lambda} \frac{d\omega}{2\pi} \frac{(\omega^2 + \omega_0^2) S_E(\omega)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2},$$

$$\frac{J_*}{K} = \frac{\gamma}{\pi\omega_0^3} \int_0^{\Lambda} \frac{\omega^3(\omega^2 + \omega_0^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} d\omega \quad (K > 0).$$

Reach stationarity first, then take $\gamma/\omega_0 \to 0$ at fixed finite $\Lambda > \omega_0$. Near ω_0, writing $u = \omega - \omega_0$, the integrand has numerator $2\omega_0^5$ and denominator $\omega_0^2(4u^2 + \gamma^2)$, and $\int du/(4u^2 + \gamma^2) = \pi/(2\gamma)$. Away from a fixed resonance neighbourhood the integral is bounded and its prefactor vanishes. Shrinking that neighbourhood afterwards proves $J_* \to K$. For any fixed even finite-cutoff spectrum continuous at resonance the same calculation gives

$$J_* \longrightarrow \frac{3\pi\epsilon_0 c^3 S_E(\omega_0)}{\omega_0^3}.$$

Thus equal limiting actions over a frequency band require a cubic spectral shape on that band. The amplitude remains free. Convergence is pointwise, with resonance inside the cutoff; dipole/nonrelativistic validity also needs small $K\omega_0/(mc^2)$. Taking $q \to 0$ at finite time removes both forcing and damping and retains initial energy, because relaxation time has diverged. It does not give the stationary-first limit.

Amplitude scaling $E \mapsto aE$ sends $(K, J_*) \mapsto (a^2K, a^2J_*)$ at fixed damping; $K = 0$ admits rest. At fixed damping, cutoff removal instead has

$$J_*/K \sim \gamma\Lambda^2/(2\pi\omega_0^3).$$

Position variance diverges logarithmically and velocity supplies the quadratic energy divergence. A joint limit needs ultraviolet estimates not supplied by narrow-resonance cancellation. This is a failure of the chosen viscous model and spectrum together, not a theorem about every causal extended-charge model. Source context: B23, Zwanzig equations (16)–(17), (23), (25)–(30), passages distinguishing friction, forcing and bath preparation. For a homogeneous isotropic transverse field, if $u(\omega)d\omega$ is the total positive-frequency energy density, isotropy and equal electric/magnetic energy give $u(\omega) = 3\epsilon_0 S_E(\omega)/\pi$. With electromagnetic mode density $\omega^2/(\pi^2 c^3)$ the chosen spectrum therefore means energy $K\omega$ per mode. Without those field hypotheses its coefficient is only a chosen one-component noise normalization.

Boyer 1969, pp. 1374–1376 and 1379–1380, passage read with formula-OCR limitations, supplies the spectral/resonance precedent and warns of the ultraviolet failure of the dipole/nonrelativistic approximation. Boyer 1975, abstract read only, explicitly supplies the random homogeneous boundary condition and its Planck normalization. The locally viscous response above and the third-order response below remain different models; neither selects a nonzero background amplitude.

Lorentz-invariant spectral form and conditional Planck calibration

A homogeneous isotropic stationary radiation background with Lorentz-invariant second-order spectrum has mode energy $\kappa\omega$, leaving the action amplitude κ free. Gaussianity is additional when Gaussian oscillator statistics are claimed. Sources: Marshall 1963, metadata; Boyer 1969, abstract. A nonzero background supplies positivity; invariance also permits zero.

In the formal third-order dipole response $M\ddot{x} = -M\omega_0^2 x + M\tau_{\rm rad}\dddot{x} + eE$, with SI $\tau_{\rm rad} = e^2/(6\pi\epsilon_0 Mc^3)$, a specified stationary Fourier response has the resonant asymptotics

$$\langle x^2\rangle_{\rm res} \sim \frac{\kappa}{M\omega_0}, \quad \langle p^2\rangle_{\rm res} \sim \kappa M\omega_0, \quad {\rm Cov}_{\rm res}(x,p) = 0, \quad \sqrt{\det \Sigma_{\rm res}} \sim \kappa.$$

The window must contain asymptotically many linewidths while remaining narrow relative to ω_0; a fixed few linewidths capture only a fraction. Linear Gaussian response is Gaussian. The squared inverse susceptibility near resonance is $4\omega_0^2(\omega-\omega_0)^2 + \tau_{\rm rad}^2\omega_0^6$, with integral $\pi/(2\tau_{\rm rad}\omega_0^4)$, yielding cancellation of charge and mass and equal kinetic/potential resonant energies. This is a band-filtered stationary-response statement, not a causal well-posedness theorem for the bare third-order equation.

Its off-resonant momentum contribution for $\omega_0 \ll \omega \ll 1/\tau_{\rm rad}$ is relatively of order $1/(\pi\tau_{\rm rad}\omega_0)$; above $1/\tau_{\rm rad}$ it diverges logarithmically. The position correction is of relative order $\tau_{\rm rad}\omega_0 \log(1/(\tau_{\rm rad}\omega_0))$. A cutoff $\Omega_c \ll \sqrt{\pi\omega_0/\tau_{\rm rad}}$ can retain resonance dominance while breaking the assumed Lorentz invariance. These ultraviolet statements concern a different response model from the stable viscous approximation above, whose energy divergence is quadratic. Goedecke 1983, metadata, and Nieuwenhuizen 2019, abstract, record the oscillator regularization issue. Access to filtered variables still needs an instrument prescription.

Conditional calibration. If an SED derivation gives the Planck thermal spectrum together with zero-point mode energy $\kappa\omega$, matching its thermal quantum gives $h_P = 4\pi\kappa$. For the further state identification $\zeta = \kappa$, this yields

$$2\zeta = \frac{h_P}{2\pi} = \frac{(\pi^4/15)^{1/3}}{2\pi} h_{\rm rad} \simeq 0.297 h_{\rm rad}.$$

At the experimental quantum identification this is $\kappa = \hbar/2$. The SED Planck derivation is disputed: Jiménez–de la Peña–Brody 1980, metadata; Senatchin 2001, passage reported in the original internal review, challenges the wall-damping argument; Boyer's 2017 review, passage as reported there, maintains a modified construction. The formula is conditional on that thermal-spectrum premise and the resonant/state identification. The radiation unit theorem itself assumes none of this SED derivation.

Nonlinear feedback: a real attractor with a supplied power scale

An effective oscillator of frequency $\omega > 0$, action $J \geq 0$ and depot energy $e \geq 0$ has the supplied laws

$$\dot{\theta} = \omega, \quad \dot{J} = 2(ge - \gamma)J, \quad \dot{e} = P - \nu e - 2ge\omega J,$$

with losses $\gamma, \nu > 0$, gain $g > 0$ of units (energy time)$^{-1}$ and power $P \geq 0$. Internal conversion cancels in total energy $E = \omega J + e$:

$$\dot{E} = P - 2\gamma\omega J - \nu e \leq P - aE, \qquad a = \min(2\gamma, \nu) > 0.$$

The positive quadrant is invariant and this proves global boundedness. At finite fuel $P = 0$, $E(t) \leq E(0)e^{-at}$ and $J \to 0$, even after initial amplification. The quadrature realization

$$J = (Q^2 + R^2)/2, \quad \dot{Q} = (ge - \gamma)Q - \omega R, \quad \dot{R} = \omega Q + (ge - \gamma)R$$

removes the angle singularity. With $x = Q/\sqrt{m\omega}$, $p = \sqrt{m\omega}R$, its energy is ωJ; gain acts on both quadratures, so this is an effective amplitude model, not unmodified Newtonian $\dot{x} = p/m$ or a closed Hamiltonian field.

For $P > P_c = \nu\gamma/g$, a positive equilibrium exists:

$$e_* = \gamma/g, \qquad J_* = (P - P_c)/(2\gamma\omega).$$

It attracts every positive seed. The nonnegative Lyapunov function

$$V = \omega[J - J_* - J_* \log(J/J_*)] + e - e_* - e_* \log(e/e_*)$$

has compact sublevel sets inside the positive quadrant. Writing deviations $\delta J, \delta e$, the oscillator term contributes $2g\omega\delta J\delta e$, while $\dot{e} = -(\nu + 2g\omega J_*)\delta e - 2g\omega e\delta J$. Multiplying by $1 - e_*/e$ cancels the cross term, giving

$$\dot{V} = -(\nu + 2g\omega J_*)\frac{(e - e_*)^2}{e} \leq 0.$$

The only invariant set with $\dot{V} = 0$ is the equilibrium, proving convergence. An empty depot becomes positive immediately at $P > 0$; an exactly zero oscillator seed remains zero even above threshold. The quiet equilibrium has action growth rate $2(gP/\nu - \gamma)$.

At the same supplied power, frequencies acquire the same energy, not the same action. A common positive K requires $P(\omega) = P_c + 2\gamma\omega K$, inserting that action into the source. Also $J_* \downarrow 0$ continuously as $P \downarrow P_c$. Attraction does not give universality or exclusion of exact rest. Mechanism source: Schweitzer–Ebeling–Tilch 1998, publisher abstract; the specific law and Lyapunov proof above are the written model calculation, without a full-text equation match or priority assessment.

Thermodynamic information costs: an inequality, not an optimum

Assume a specified measurement/reset thermodynamic account requires mean work $W \geq k_B T\, I$ for gained mutual information I in nats. The conditions of that cycle, system free-energy changes and memory preparation are part of this premise; it is not asserted for every possible measurement/erasure process. The cited ingredients are Landauer 1961, Bennett 1982 and Sagawa–Ueda 2009, metadata.

For a uniform prior on area A_0, if almost every posterior is supported on area at most η, its entropy is at most $\log \eta$ while the prior entropy is $\log A_0$. Hence $I \geq \log(A_0/\eta)$ and

$$W \geq k_B T \log(A_0/\eta), \qquad \eta \geq A_0 e^{-W/(k_B T)}.$$

The area unit cancels. This is a necessary bound, not a proof that the minimal attainable area equals its right side. Finite-bit storage of a finite-precision reading does not bound every preparation, control and reset cost, nor establish equality in the stipulated work law. Theorem I supplies arbitrarily fine finite classical protocols under its idealized resource access; the inequality by itself adds no universal positive resolution floor when the work budget is unrestricted. Identifying $k_B T\tau$ with an action unit would additionally fix a time scale that this information bound never chooses.

Consequence for STATE

The radiation integral fixes an action unit under its stated equilibrium assumptions. Resonant response, a pumped attractor, classical posterior inference and thermodynamic accounting impose different conditional laws. None bridges all four obligations: classical dynamics, positive record floor, quantum states/instruments, and a universal scale identified with physical $\hbar$. The exact zero-field, finite-fuel, squeezing and complete-record countermodels remain admitted wherever their stated operations are allowed.